

CO6-AT2

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Operating Systems

2) File Allocation - Cloud Storage

→ The three main allocation methods are contiguous, linked, and indexed allocation.

→ Contiguous allocation: Stores all blocks of a file next to each other. It provides very fast sequential and direct access because fewer disk movements are required.

→ Linked allocation: File blocks can be stored anywhere, with each block pointing to the next. It is good for sequential access but slow for direct access.

→ Indexed allocation: Uses an Index block containing pointers to all file blocks. It supports direct access but requires additional space and an extra access to the index.

Best Choice: Contiguous Allocation.

→ For a cloud storage company where the main problem is frequent file access delays, contiguous allocation is most suitable because -

1. Blocks are stored continuously.
2. Data access time is reduced.
3. Direct access is faster.

Disk Scheduling - Data Centre

→ The three algorithms mentioned are FCFS, SSTF and SCAN.

• FCFS (First Come First Serve) -

Processes requests in the order they arrive. It is simple but can cause large head movements.

• SSTF (Shortest Seek Time First) -

Select the request closest to the current head position.
It reduces immediate seek time but can cause starvation.

• SCAN -

→ The disk head movement in one direction, servicing requests along the way, then reverses direction.

Best choice: SCAN

→ SCAN is appropriate for the data centre because:

1. It significantly reduces unnecessary head movement.
2. Requests are serviced systematically.
3. It provides better average seek time than FCFS.
4. It reduces the possibility of starvation compared with SSTF.
5. It performs well under heavy workload.

→ SCAN provides a good balance between low seek time and fair request servicing, making it suitable for a busy data centre.

Course

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Unit 2: Data Storage
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Unit 4: Data Security
Unit 5: Data Archiving
Unit 6: Data Migration
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30 Bitmap Free Space Management

→ A bitmap uses one bit for each disk block to indicate whether the block is free or allocated. As the disk becomes larger, scanning the entire bitmap can become slow. The problem also for an optimization technique to improve allocation efficiency.

Proposed technique: Hierarchical Bitmap

Instead of searching the complete bitmap every time:

1. Divide the bitmap into smaller groups.
2. Maintain a higher-level bitmap indicating which groups contain free blocks.
3. First search the higher-level bitmap.
4. Identify a group containing free space.
5. Search only that smaller bitmap for the required block.

Advantages:

- Reduces bitmap scanning time.
 - Makes allocation faster for larger disks.
 - Requires fewer memory accesses.
 - Scales better as disk size increases.
 - Efficiently identifies free blocks.
- A hierarchical or multi-level bitmap is an effective optimization because it avoids scanning the entire bitmap for every allocation.

4) SCAN is C-SCAN for Real-Time Applications

→ The application requires consistent and predictable disk access time. The question arises whether SCAN or C-SCAN is more suitable.

SCAN:

→ SCAN moves the disk head in one direction and services requests, then reverses direction.

Advantages:

- + Good overall performance.
 - + Reduces starvation.
 - + Efficient for heavy workloads.
- However, requests near the middle of the disk may receive better service than requests at the edges.

C-SCAN:

→ C-SCAN moves in one-direction only. After reaching the end, the head returns directly to the beginning without servicing requests during the return.

Advantages:

1. Provides more uniform waiting time.
2. Gives requests a more predictable service platform.
3. Provides fairness across the disk.
4. Suitable for systems requiring consistent access time.

Best Choice: C-SCAN

→ For a real time application, C-SCAN is more suitable because it provides more uniform and predictable waiting times.

- + Choose C-SCAN when predictability and consistency are more important than simply minimizing average seek time.

G) Hybrid Storage System:-

→ The company wants to combine file allocation and disk scheduling techniques for large-scale data access.

Recommended Combination: Contiguous Allocation + SCAN.

Contiguous Allocation: → Is suitable for storing frequently accessed large file because:

- + File blocks are placed close together.
- + Sequential access is very fast.
- + Disk-head movement is reduced.
- + Disk access is efficient.

SCAN → is suitable for scheduling disk requests because:

- + It reduces unnecessary head movement.
- + Requests are processed systematically.
- + It provides fairness.
- + It performs well under heavy workloads.

How they work together.

User request → Contiguous File Blocks → SCAN disk

Scheduling → Data Retrieval.

For example, if a large database file is stored in consecutive blocks, SCAN can efficiently service requests from different users while controlling disk-head movement.

Performance benefits:-

Factor	Benefit -
Access speed	High due to contiguous block.
Seek time	Reduced by SCAN.
Throughput	Improved.
Fairness	Better request servicing
Large files	Efficient sequential access
Overall performance	High.

Contiguous allocation + SCAN is a practical combination for a large scale storage system because it combines fast file access with efficient and fair disk scheduling.